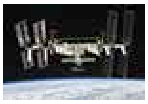
Antarctic ice shelf develops huge crack
NASA scientists have a noticed an iceberg twice the size of New York city that is about to break off. **https://dgit.in/mar19-23**



X-rays for space communications
NASA says X-rays, because of shorter wavelengths than both radio waves and lasers, might be better. **https://dgit.in/mar19-24**

distribute the receiving telescopes. However, practical considerations such as the physical construction and linking of the telescopes with cables would make the random distribution very difficult, and expensive. A suitable compromise between cost and resolution, was the spiral configuration chosen.

Credit: SKA organisation

The layout of the SKA

The design approach allows for researchers to use the entire array as one single telescope, or divide the dishes up into multiple smaller telescopes, by smartly processing the signals from an arbitrary subset of the SKA. Now the distances between the dishes are known, and this is factored in by computers that process the signals. Whether the entire array is working as a single telescope or multiple smaller telescopes, the principle is the same. The signals from multiple dishes are combined together and synthesised by computers, effectively creating an astronomical instrument that stretches the distance between the telescopes receiving the signals.

Persistent Systems, a company based in Pune, is tasked with designing the software for controlling the telescope, as well as managing the data. Snehal Valame, engineering head at Persistent Systems, explains the sheer volume of the information that will be gathered, and how it will be processed “here’s a fact to give you a sense of just how much data the SKA is expected to collect: a single day’s worth of collected data would take around 2 million years to play back on an iPod. We’re gearing up to take on that huge volume of data by willing to extend our expertise in the area like Big Data and Machine Learning. SKA currently is in its bridging phase, moving from pre-construction to the construction phase Activities like laying out an efficient design, software architecture, minimizing the risks for construction period via prototyping for all the SKA consortia are in progress. Persistent Systems is involved in two consortia under SKA: Telescope Manager, and Signal and Data Transport. Telescope Manager deals with the Telescope Controls and Observation Management for SKA, and Signal and Data Transport deals with designing a resilient and robust monitoring network system as is required for the heterogeneous network of SKA. We will be drawing from the prototyping we did for The GMRT.” The central computer of the SKA will have the processing power of one hundred million desktop computers.

THE SCIENCE GOALS

There are several science goals of the SKA, each of which have their own working group. Our telescopes peered into distances of 13 billion light years from Earth, at a time when the universe was less than one billion years ago. The SKA will be able to peer farther back into time, and witness the formation of the first galaxies. This was when the first proto galaxies and quasars were being formed, the faint light from these objects are absorbed by other matter between these objects and the telescopes of the Earth. The SKA will be able to study the objects with extreme redshifts, and produce high res images of these radio sources. These are expected to be images of the first light sources in the universe. The SKA will also be able to scan the skies of exoplanets, looking for biosignatures, or molecules and elements that signify the presence of life, including amino acids. The telescope will also investigate planetary formation in accretion disks around young stars. The SKA will also explore the mysterious phenomenon of acceleration, trying to understand why the universe is expanding faster, and by how much.

Another mysterious phenomenon the SKA will explore, is that of Dark Energy, an unknown force that seems to counteract the phenomenon of gravity. This is the force that makes the galaxies fly apart from each other, instead of being attracted to each other. The sensitivity of the telescope will allow researchers to investigate magnetic fields of other stars and planets, which are invisible to optical telescopes. The SKA will do so by observing the interactions of high energy particles with these magnetic fields. The SKA is being built and designed as a sensitive and versatile discovery machine, flexible enough to find things that scientists have not yet anticipated.

Credit: SKA organisation

An artist’s impression of the SKA array in South Africa